Question. A hotel chain reports that the length of stay per reservation has a mean of 2.8 nights and a standard deviation of 1.9 nights. The distribution of lengths of stay is skewed to the right. A revenue manager randomly selects 40 reservations. Find the probability that the mean length of stay in the sample is greater than 3.2 nights. Show all steps using the SPDC structure.

Detailed prompts

S – State:

1. Define the population (true) parameter in context.
2. Define the sample parameter in context.
3. What is 40? What does it represent?
4. What are 2.8 nights and 1.9 nights? What do they represent? What is the shape of the population distribution?
5. Write the problem (the probability you are asked to find using $\bar{x}$).

P – Plan:

1. Check randomness: Was the sample random? Why does randomness matter?
2. Check independence (10% condition): How many reservations must there be in the population? Is $n < 10\%$ of N?
3. Check normality: The population is skewed. Is the sample size large enough for the Central Limit Theorem ($n \geq 30$)? What is the shape of the sampling distribution of $\bar{x}$?
4. Compute the mean of the sampling distribution of $\bar{x}$.
5. Compute the standard deviation of the sampling distribution of $\bar{x}$, and write the Normal model.

D – Do:

1. Write the sampling distribution of the sample statistic.
2. Write the probability statement in terms of $\bar{x}$.
3. Write the probability statement in terms of Z formula.

4. Convert the boundary to a z-score.

5. Use the standard Normal table and write the value of the probability.

C – Conclude:

1. Start your conclusion with "There is about a ___ probability that…"

2. What is the sample size, and what is in the sample?

3. What statistic does the sample produce?

4. What value is in the question?

5. What does this mean describe in context?